A PROJECT TITLE

ON

INSIDER THREAT DETECTION USING ELASTIC NET AND RANDOM FOREST

Project report submitted as a partial fulfillment of the requirements for the course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

MACHINE LEARNING

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DEPARTMENT OF:

Name of Title: INSIDER THREAT DETECTION USING ELASTIC NET AND RANDOM FOREST

1. ABSTRACT

Insider threats are security risks caused by people who already have authorized access to an organization's computer systems. Detecting such threats is difficult because suspicious activities can sometimes look similar to normal user behavior. This project uses machine learning to identify unusual activities and classify them as normal or potentially malicious. Elastic Net is used to select important features from user activity data, while Random Forest is used to classify the behavior. The proposed approach can help organizations detect suspicious activity at an early stage and reduce the risk of data theft, misuse of systems, and other security incidents.

2. INTRODUCTION

Cybersecurity is important for protecting an organization's systems and information. An insider threat occurs when an employee or other authorized user misuses access to company resources. Traditional rule-based monitoring may not detect every unusual pattern. Machine learning can learn from historical user activity and identify patterns associated with normal and suspicious behavior. In this project, Elastic Net helps reduce unnecessary features and Random Forest learns from the selected features to predict whether new activity is normal or a potential insider threat.

3. SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux
- Programming Language: Python
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib
- Development Environment: Jupyter Notebook / VS Code
- Dataset: Insider-threat or user-behavior activity dataset

Signature of the Guide **Course Instructor**